TEXORA · SESSIONAL NOTES · EXPERIMENT SHEET

Rot-proofing, Soil Release, Antistatic and Antibacterial Finishes

Four small protective productions — stop microbes rotting the fibre, let soil wash out of polyester, drain static charge, and keep bacteria off the cloth.

WP-4202 · TEXTILE FINISHING SESSIONAL · EXPERIMENT 2 · SPECIALIZATION — WET PROCESSING

1. Aim and theory

Aim: to carry out four small pad-applied protective finishes, each on the fibre it is designed for — (A) **rot-proofing** of cotton, (B) **soil-release** finish on polyester, (C) **antistatic** finish on polyester, and (D) **antibacterial** finish on cotton — recording a working recipe, an add-on for each, and a first assessment note. The four finishes look similar on the bench (all are padded from water) but solve four chemically different problems, and confusing them is a classic examination error.

(A) Rot-proofing. Cellulosic fabrics buried in soil, used as tents, awnings, sandbags or geotextiles, or stored damp are digested by cellulase enzymes from bacteria and fungi. The fabric loses strength long before it looks destroyed — the trade calls the loss *tendering*. Rot-proofing deposits a microbial poison in/on the fibre so organisms cannot colonise it. The classical agents are metallic: copper naphthenate or copper 8-hydroxyquinolinolate (green, effective, used for decades on military canvas), and chromium–copper systems related to timber preservation. Their weakness is colour, metal leaching and modern effluent limits. Modern teaching practice replaces them where possible with organic biocides — e.g. zinc pyrithione, isothiazolinones, or bonded quaternary systems — and, for many end uses, by choosing a rot-resistant synthetic fibre instead. In this sessional the copper-based recipe is demonstrated at bench scale because its chemistry (and its green colour) makes the deposition visible, with the modern alternatives taught alongside.

(B) Soil release. Polyester is hydrophobic and oleophilic: oily soil spreads on it, anchors strongly, and ordinary washing — which works by water lifting soil from a hydrophilic surface — fails. A soil-release finish applies a *hydrophilic polymer* (typically a PEG–terephthalate block copolymer, chemically a cousin of the polyester fibre itself, so it anchors durably) that stays invisible when dry but swells in the wash liquor, presenting a water-loving layer between fibre and soil. The soil then rolls up and detaches. The finish does not stop soiling; it makes soiling reversible. It is the standard finish on polyester workwear and table linen.

(C) Antistatic. Synthetic fibres are electrical insulators with very low moisture regain, so friction (in processing, in wear, in tumble drying) builds static charge that causes cling, dust attraction, sparking and handling shocks. An antistatic finish provides a leakage path: hygroscopic and/or ionic agents — classically quaternary ammonium compounds, also poly(ethylene glycol) esters and ethoxylates — attract a thin moisture film and conduct charge away. Topical antistats are effective but wash off; durable versions use crosslinkable or fibre-reactive systems, and the permanent industrial solution is a conductive fibre blended into the fabric. Performance is humidity-dependent: every topical antistat looks worse at 25% relative humidity than at 65%.

(D) Antibacterial. An antibacterial (antimicrobial) finish inhibits or kills bacteria on the fabric, controlling odour from sweat decomposition and protecting hygiene textiles. Two families must be distinguished: *leaching* agents (e.g. triclosan historically; silver-ion release systems) migrate out and create a zone of inhibition around the fabric but deplete and enter the effluent; *bonded* agents (e.g. silane–quaternary ammonium compounds, Si-Quat type) are fixed to the fibre, kill on contact, and do not leach — they show no zone of inhibition, which is a property, not a failure. Polyhexamethylene biguanide (PHMB) is a widely used intermediate case. Quantitative proof of antibacterial action (AATCC 100 % reduction) is worked in Experiment 4.

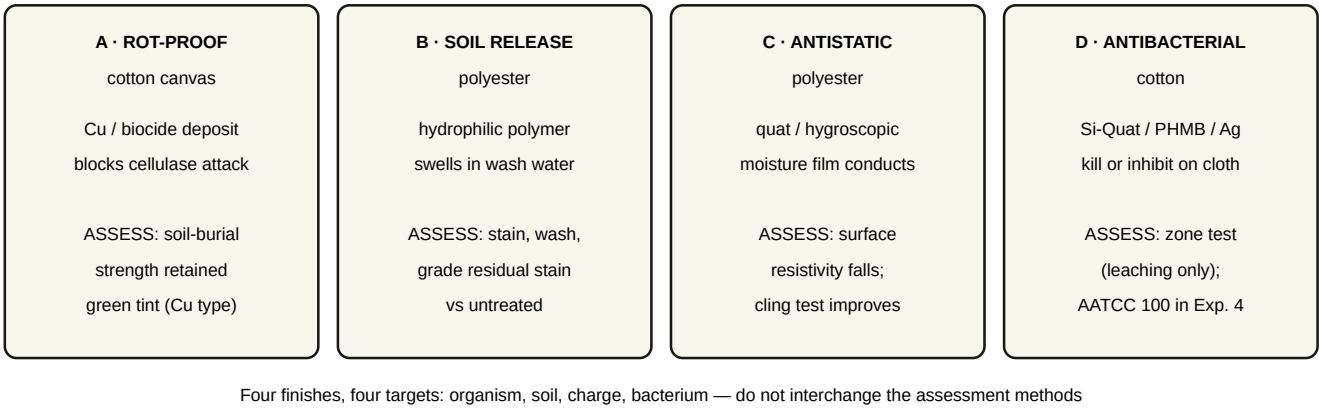


Fig. 1 — The four productions of this experiment: agent, mechanism and first assessment for each panel.

2. Apparatus, materials and safety

ITEM	SPECIFICATION / USE
Laboratory padder, stenter/oven	As Experiment 1; all four finishes padded at 70% wet pick-up unless stated
Fabric	Cotton canvas / poplin for A and D; polyester poplin for B and C; pieces about 20 cm × 15 cm

Chemicals A	Copper naphthenate emulsion (or copper 8-hydroxyquinolinolate dispersion); modern alternative for discussion: zinc pyrithione dispersion
Chemicals B	Soil-release polymer (PEG–terephthalate copolymer emulsion); mild alkali for bath pH
Chemicals C	Quaternary ammonium antistatic agent; alternative: PEG-ester non-ionic antistat
Chemicals D	Antibacterial agent — Si-Quat type (bonded) with PHMB type as comparison; binder-free
Assessment kit	Standard oily soil / used engine oil + carbon black for staining; wash bath with 2 g/L detergent; multimeter with surface electrodes or cling-test rig; nutrient agar plates only under microbiology supervision

Safety. Biocides are poisons by design: gloves for all four baths, no mouth pipetting ever, and treated fabrics for the antibacterial panel are handled as microbiology samples once inoculated — that part is demonstrated/supervised, not free bench work. Copper baths and washings go to heavy-metal waste, never to the sink. Label every treated piece on both sides of the record card: an unlabelled biocide-treated fabric must not enter general use.

3. Production A — rot-proofing of cotton

Recipe A (copper type, bench demonstration)

COMPONENT	AMOUNT	FUNCTION
Copper naphthenate emulsion (as supplied, about 8% Cu)	50 g/L	Rot-proofing agent; target about 0.5–0.8% Cu on fabric weight
Wetting agent	1 g/L	Penetration of heavy canvas
Wet pick-up	70%	Add-on of emulsion = 50 × 0.70 = 35 g/kg (3.5% owf)

$$\text{Emulsion add-on} = 50\ \text{g/L} \times 0.70 = 35\ \text{g/kg fabric}$$

1. Pad the scoured cotton canvas twice, dry at 110 °C. A green tint develops — with copper systems, colour is the visible proof of deposition (and their commercial limitation for apparel).
2. No high cure is needed for the naphthenate emulsion; the metal soap deposits in the fibre lumen and on the surface as the water leaves. Fixation improves with a subsequent light resin treatment in industrial practice.

3. **Modern note:** repeat mentally with a zinc pyrithione or isothiazolinone product at its supplier dose (typically 10–30 g/L): colourless and drain-friendlier, but with narrower spectrum and questions over long-term outdoor durability. State in the record which family was actually used.
4. **Assessment note:** the reference method is a *soil-burial test*: strips are buried in active compost/soil for a set period (commonly 7–28 days), exhumed, washed and tensile-tested. Untreated cotton can lose most of its strength in days; a successful rot-proof keeps strength loss small. A same-day bench proxy is simply to record colour, handle and add-on, and to set up the burial comparison for later breaking — never claim rot-proofing from appearance alone.

4. Production B — soil-release finish on polyester

Recipe B

COMPONENT	AMOUNT	FUNCTION
Soil-release copolymer emulsion	30 g/L	Hydrophilic film; terephthalate blocks anchor to polyester
NaOH (dilute, to pH)	pH 8.0–8.5	Optimum bath stability / deposition for most SR polymers
Wet pick-up	70%	Add-on = 30 × 0.70 = 21 g/kg (2.1% owf)

$$\text{Add-on} = 30\ \text{g/L} \times 0.70 = 21\ \text{g/kg fabric}$$

1. Pad polyester twice, dry at 110 °C, then heat-set/cure at 170–180 °C for 45–60 s: the copolymer's polyester-like blocks co-crystallise with the fibre surface, which is the source of its wash durability.
2. **Assessment:** stain treated and untreated pieces identically (a drop of used engine oil + a carbon-black/mineral-oil standard stain, aged 1 h), then wash both together at 40 °C with 2 g/L detergent for 15 min, rinse and dry. Grade the residual stain against the photographic soil-release rating scale (AATCC 130 principle, grades 1 = severe residual stain to 5 = no visible stain). A working finish typically lifts the grade by 2–3 points against the untreated control washed in the same bath — the shared bath is what makes the comparison fair.
3. Record also that the dry treated fabric feels and looks like untreated polyester: the hydrophilic layer only expresses itself in water.

5. Production C — antistatic finish on polyester

Recipe C

COMPONENT	AMOUNT	FUNCTION
Quaternary ammonium antistat	15 g/L	Ionic, hygroscopic conductive layer
Wet pick-up	70%	Add-on = 15 × 0.70 = 10.5 g/kg (1.05% owf)

$$\text{Add-on} = 15\ \text{g/L} \times 0.70 = 10.5\ \text{g/kg fabric}$$

1. Pad polyester twice, dry at 100–110 °C. No cure is required for a topical quat; a durable version would add a crosslinker and cure, at some cost to handle.
2. Condition all pieces at standard atmosphere (20 ± 2 °C, 65 ± 4% RH) for 4 h before testing — antistatic results quoted without humidity are meaningless.
3. **Assessment:** (i) surface resistivity with two parallel electrodes and a high-resistance meter: untreated polyester typically reads above 10^{12}–10^{13} Ω; a working topical antistat brings the reading down by roughly two to three orders of magnitude (state the measured values, not the textbook ones, in the record). (ii) Cling test: rub the fabric with a standard material and observe clinging to a metal plate / attraction of ash or paper bits — treated fabric sheds charge in seconds, untreated clings for minutes. (iii) Note durability honestly: wash one treated piece five times and re-test; a topical quat loses most of its effect, which is exactly why conductive-fibre blends exist.

6. Production D — antibacterial finish on cotton

Recipe D

COMPONENT	AMOUNT	FUNCTION
Antibacterial agent, Si-Quat type (as supplied)	20 g/L	Bonded contact-kill agent; silane end anchors to cellulose
Bath pH	pH 5.0–6.0	Hydrolysis/condensation window for the silane bond
Wet pick-up	70%	Add-on = 20 × 0.70 = 14 g/kg (1.4% owf)

$$\text{Add-on} = 20\ \text{g/L} \times 0.70 = 14\ \text{g/kg fabric}$$

1. Pad cotton twice, dry at 110 °C, cure briefly at 130–150 °C for 2 min to complete silane condensation with fibre –OH groups.
2. **Assessment note 1 — zone of inhibition (qualitative):** a disc of treated fabric on inoculated agar. A *leaching* agent (PHMB/silver type) shows a clear halo where bacteria cannot grow. The bonded Si-

Quat shows **no halo**, with growth stopped only under the disc — record this as the expected signature of a non-leaching finish, not as failure.

3. **Assessment note 2 — quantitative:** the industry number comes from AATCC 100 (inoculate, incubate 24 h, recover and count colonies; report % reduction). The fully worked example, with real arithmetic, is set out in Experiment 4, Section 5, using this production's fabric type.
4. Durability note: bonded systems are specified precisely because they survive laundering (claims of 25–50 washes are common commercially and must be verified, not repeated, in the record). Leaching systems deplete — the same halo test after 10 washes shrinks visibly.

7. Results (as recorded — template with typical values)

PRODUCTION	ADD-ON	FIRST ASSESSMENT
A — rot-proof, cotton	35 g/kg emulsion (3.5% owf)	Even green tint; burial strips set up for later strength test
B — soil release, polyester	21 g/kg (2.1% owf)	Residual-stain grade improved by about 2–3 grades vs control in same wash
C — antistatic, polyester	10.5 g/kg (1.05% owf)	Surface resistivity down by about 2–3 orders of magnitude at 65% RH; cling released in seconds; effect largely lost after 5 washes
D — antibacterial, cotton	14 g/kg (1.4% owf)	Si-Quat: no zone, growth inhibited under disc only; quantitative test deferred to Exp. 4

> **EXAMINER'S DISTINCTION**
>
> Rot-proofing protects *the fabric* from organisms. Antibacterial finishing protects *the wearer / the product environment* from bacteria on the fabric. The chemistries overlap (both use biocides) but the test methods and the claims allowed are different — say which one you mean, every time.

8. Precautions

- Keep the four baths strictly separate and label troughs: a trace of cationic quat (C) can break an anionic emulsion in another bath.
- Condition antistatic samples before testing and record temperature and RH with every resistivity value.
- Wash soil-release treated and control pieces in the *same* bath; separate baths invalidate the comparison.

- Copper waste and biocide washings to designated waste only. Treated fabrics stay labelled until disposal or verified testing.
- Never extrapolate a halo test to a % kill, or a % kill to a health claim. Each assessment answers only its own question.

9. Practice (viva preparation)

1. Why does rot show itself as strength loss before the fabric looks damaged, and which test proves rot-proofing?
2. A soil-release finish does not prevent soiling. Explain, with the role of water, how it works in the wash.
3. Why do topical antistatic results depend on relative humidity, and why does washing remove the effect of Recipe C?
4. A treated fabric shows no zone of inhibition on agar. Give two opposite interpretations and state how you would decide between them.
5. Calculate the add-on for Recipe D if pick-up is raised to 80%, and state one reason a finisher might still keep 70%.
6. Why are copper rot-proofing systems now restricted for apparel, even though they work well?

Answer key

1. Cellulase enzymes cut cellulose chains throughout the fibre (falling degree of polymerisation = falling tensile strength) long before mass loss or holes appear. Proof is retained tensile strength after a standard soil-burial exposure, treated vs untreated.
2. The dry polymer film is inconspicuous; in wash liquor it hydrates and swells, so the fibre–soil interface becomes water-rich. Detergent solution undercuts and rolls up the oily soil from the hydrophilic layer, whereas on bare polyester the oil wets and anchors directly to the fibre.
3. The conductive path is a moisture film attracted by the hygroscopic/ionic agent; at low RH the film thins and resistivity rises sharply. The quat in Recipe C is only physically deposited, so laundering dissolves and removes it — durability needs crosslinking or a conductive fibre.
4. (i) The agent is bonded/non-leaching and kills only on contact — no halo is expected. (ii) The agent is absent or inactive — nothing is there to leach. Decide with a quantitative contact test (AATCC 100, Exp. 4): case (i) gives a high % reduction, case (ii) gives about zero.
5. Add-on = 20 × 0.80 = 16 g/kg (1.6% owf). Reasons to stay at 70% include cost, handle, bath stability at the nip, and avoiding excess unbound agent that would wash off in first use.
6. Colour (green tint), metal content in effluent and on skin, and modern chemical regulations/customer restricted-substance lists. Effectiveness was never the problem; acceptability

is.